CSA1718 : Artificial Intelligence

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3.Water Jug Problem:

```
from collections import deque

def solve():
    queue = deque([((0, 0), [])])
    visited = set()

    while queue:
        (a, b), path = queue.popleft()

        if (a, b) in visited:
            continue
        visited.add((a, b))
        path = path + [(a, b)]
```

```
        if a == 2:
            return path

        x = min(a, 3 - b)
        y = min(b, 4 - a)

        states = [
            (4, b), (a, 3), (0, b), (a, 0),
            (a - x, b + x), (a + y, b - y)
        ]

        for s in states:
            if s not in visited:
                queue.append((s, path))

for state in solve():
    print(state)
```

Output:

```
(0, 0)
(4, 0)
(1, 3)
(1, 0)
(0, 1)
(4, 1)
(2, 3)
```